

CHAPTER 2

UNATTENUATED RAYLEIGH WAVE IN HOMOGENEOUS ELASTIC HALF-SPACE

Introduction.

Earthquakes damage to civil engineering structures is heavily influenced by both local soil properties and other factors, such as irregular surface topography and underground morphology. The H/V-method is considered one of the most effective approaches to characterize soil properties. This method provides an estimation of the ratio between the Fourier amplitude spectra of the horizontal (H) and vertical (V) components of ambient noise vibrations recorded at a single station (Tuan, T. T., 1980).

Given that surface waves are known to be the primary contributors to ambient vibrations, although their composition may vary across different sites, the objective of this thesis is to conduct a thorough investigation into the behavior of the H/V-ratio of Rayleigh waves and its variants, e.g. Lamb waves. This research aims to contribute to the further advancement of the established H/V-method. The approach outlined in this thesis involves studying the H/V-ratio of surface waves, progressing from simpler to more complex models. The range of model complexity spans from the easiest model, known as the homogeneous half-space, to more complex models such as multilayered homogeneous media with rigid and flexible bottom boundaries.

An important phenomenon worth examining is that soil functions as a material that lies between the state of a solid and a liquid. In the case of a soft soil layer situated atop a hard halfspace, it possesses the ability to transmit prograde Rayleigh waves (Tanimoto et al., 2005; Malischewsky Auning et al., 2006, 2008; Hernandez-Aguirre et al., 2019). When subjected to intense earthquake activity, the rigidity of soils diminishes, and their Poisson's ratio can approach 0.5, resembling that of a liquid. Gaining an understanding of the prograde particle motion in such circumstances becomes crucial for assessing the potential damage incurred by soft ground.

In earthquake engineering, the stability of slender buildings, equipment and instruments in essential facilities and lifelines is extremely important. When constructing structures, they are designed to bear vertical loads exceeding their own weight. Consequently, the impact of the vertical component of earthquake loads is often overlooked. However, vertical loads can pose a dual threat to buildings and other structures. Firstly, through the peak acceleration of the ground,

and secondly, through the P- Δ effect. The latter becomes significant when the ground tilts in the direction of propagation, causing an increase in the overturning moment. In simpler terms, when particle motion follows a prograde path, the P- Δ effect aligns with the horizontal ground acceleration, possibly resulting in equipment and instruments toppling over. The stability of rigid bodies, representing equipment and instruments in essential and lifelines facilities, will be the subject of the end chapter of this thesis.

In this chapter, we will investigate Rayleigh waves using the simplest model known as the homogeneous half-space. This model has been extensively studied in the past. Upon analyzing seismograms, seismologists discovered a new type of wave with a velocity lower than that of transverse waves. This wave is characterized by its surficial nature, with its oscillation amplitude decreasing significantly as it travels deeper. Additionally, the dissipation of energy associated with this wave is slower when compared to longitudinal and transverse waves. The latter two waves propagate in all directions from the epicenter, like during an earthquake, while the surface wave remains confined to the surface and propagates in a two-dimensional space. Hence, we can consider it as a plane wave, far away from the source center. In 1885, Lord Rayleigh predicted the existence of these elastic surface waves based on theoretical calculations.

Another important topic of interest, due to its effect on the interpretation of ground motion records, is the one related to the complex roots of the Rayleigh equation. The complex roots of the Rayleigh wave equation have often been dismissed by many authors as unrealistic or the complex roots have been considered as non-physical (Graff, 1975, Achenbach, 1973; Miklowitz, 1978). However, it is known that these complex roots arise for Poisson ratios above 0.263 (Schroder et al., 2001). These authors claim that the real part of these complex roots represents wave phase velocities that are always faster than the Rayleigh wave phase velocity but slower than the compressional wave velocity. On the other hand, Schroder et al. point out that the imaginary part of the complex roots reflects the decaying amplitude of the wave as it moves away from the source in horizontal directions. They also mention that these inhomogeneous waves, known as leaky waves, exhibit characteristics such as prograde particle elliptical motion at the free boundary, particularly for high Poisson ratios. However, in this chapter, it is shown that those roots mentioned by Schroder are not intrinsic to the Rayleigh period equation. These roots are a result of an unnecessary algebraic manipulation intended to solve the Rayleigh period equation.

In this chapter some important properties of Rayleigh waves are demonstrated:

1. Many books (e.g., Ewing, W. M., et al., 1957; Achenbach, J.D., 1973; Stein, S., and Wysession, M., 1979; Ben-Menahem, A., and Singh, S. J., 1980; Pujol, J., 2003), dissertations (e.g., Tuan Tran, 2009), and research papers (e.g., Schroder et al., 1961) that deal with the derivation and solution of the Rayleigh equation have traditionally rationalized the equation by squaring it to find its roots. This process leads to the identification of three real roots for Poisson ratios smaller than 0.263, and one real root plus two complex roots for Poisson ratios larger than 0.263. In contrast, the present thesis chapter demonstrates that there is no need of rationalization to solve the Rayleigh Period Equation. The Rayleigh Period Equation can be directly solved, yielding two real roots: one zero root and one real positive root, consistently satisfying the condition that the Rayleigh wave phase velocity is smaller than the shear wave phase velocity.
2. Although in the literature related to Rayleigh waves the elliptical retrograde motion at the free boundary is mentioned, this assertion has not been explicitly formulated by some authors. In this chapter the equation of the elliptical motion is derived.
3. In most books (Stein, S. and Wysession, M, 1979 [5]; Aki, K., Richards, G., 2009 [6]) the change of particle motion from retrograde to prograde is described. In this chapter this change in motion is numerically shown. Also, it is shown that this change in motion occurs at a depth approximately equal to one-fifth of the wavelength, and it moves closer to the surface as the Poisson ratio value increases.
4. Some researchers (Lomnitz, C., 2004 [3]; Malischewsky, P. G., 2006 [4]) have suggested the existence of prograde elliptical motion of Rayleigh waves at the ground surface. In this chapter it is shown that for Rayleigh wave propagating in homogeneous media, the particle motion at ground surface or free boundary is always retrograde for real roots of the wave equation.

The Wave Equation

It would be helpful to briefly revisit the formulation of the wave equation in an isotropic, linear elastic medium.

The equation of dynamic equilibrium of forces or equation of motion for elastic wave propagation absent of body forces (far from the source) in an elastic and homogeneous medium, is given by the homogeneous equations:

$$\sigma_{ij,j}(x,t) = \rho \frac{\partial^2 u}{\partial t^2} \quad (2.1)$$

where $\sigma_{ij,j}$ stands for the divergence of the stress tensor.

The equation of motion can be expressed in terms of displacements by utilizing the medium's constitutive relation between stress and strain tensors, as well as the strains derived from displacement. It is important to note that while the equation of motion itself is independent of the elastic constants, the solutions rely on these constants and are also influenced by spatial and temporal factors. In Cartesian coordinates (x_1, x_2, x_3) , the complete equation of motion is formulated as follows.

$$\begin{aligned} \frac{\partial \sigma_{x_1 x_1}}{\partial x_1}(x,t) + \frac{\partial \sigma_{x_1 x_2}}{\partial x_1 \partial x_2}(x,t) + \frac{\partial \sigma_{x_1 x_3}}{\partial x_1 \partial x_3}(x,t) &= \rho \frac{\partial^2 u_1}{\partial t^2}(x,t) \\ \frac{\partial \sigma_{x_2 x_1}}{\partial x_2 \partial x_1}(x,t) + \frac{\partial \sigma_{x_2 x_2}}{\partial x_2}(x,t) + \frac{\partial \sigma_{x_2 x_3}}{\partial x_2 \partial x_3}(x,t) &= \rho \frac{\partial^2 u_2}{\partial t^2}(x,t) \\ \frac{\partial \sigma_{x_3 x_1}}{\partial x_3 \partial x_1}(x,t) + \frac{\partial \sigma_{x_3 x_2}}{\partial x_3 \partial x_2}(x,t) + \frac{\partial \sigma_{x_3 x_3}}{\partial x_3}(x,t) &= \rho \frac{\partial^2 u_3}{\partial t^2}(x,t) \end{aligned} \quad (2.2)$$

where ρ is the mass density.

To express the equation of motion in terms of displacements, the constitutive law for an elastic isotropic material is used, and is given as,

$$\sigma_{ij} = \lambda \theta \delta_{ij} + 2\mu e_{ij} \quad (2.3)$$

where λ and μ are the Lamé coefficients, δ_{ij} is the Kronecker Delta, e_{ij} is the strain tensor, and θ is the dilatation, which can be written as,

$$\theta = e_{ii} = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} = \nabla \cdot \mathbf{u} \quad (2.4)$$

and the vector operator, $\nabla = i_1 \frac{\partial}{\partial x_1} + i_2 \frac{\partial}{\partial x_2} + i_3 \frac{\partial}{\partial x_3}$.

Then, the stresses can be written in terms of strains, as follows.

$$\begin{aligned} \sigma_{x_1 x_1} &= \lambda \theta + 2\mu \frac{\partial u_1}{\partial x_1} = \lambda \theta + 2\mu \frac{\partial u_1}{\partial x_1} \\ \sigma_{x_1 x_2} &= 2\mu \left[\frac{1}{2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) \right] = \mu \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) \\ \sigma_{x_1 x_3} &= \mu \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right) \end{aligned} \quad (2.5)$$

Now, taking partial derivatives in (2.5) yields,

$$\begin{aligned} \frac{\partial \sigma_{x_1 x_1}}{\partial x_1} &= \lambda \frac{\partial \theta}{\partial x_1} + 2\mu \frac{\partial^2 u_1}{\partial x_1^2} \\ \frac{\partial \sigma_{x_1 x_2}}{\partial x_2} &= \mu \left(\frac{\partial^2 u_1}{\partial x_2^2} + \frac{\partial^2 u_2}{\partial x_2 \partial x_1} \right) \\ \frac{\partial \sigma_{x_1 x_3}}{\partial x_3} &= \mu \left(\frac{\partial^2 u_1}{\partial x_3^2} + \frac{\partial^2 u_3}{\partial x_3 \partial x_1} \right) \end{aligned} \quad (2.6)$$

By plugging (2.6) into the left side of the first equation (2.2), it is obtained,

$$\begin{aligned} \frac{\partial \sigma_{x_1 x_1}}{\partial x_1} + \frac{\partial \sigma_{x_1 x_2}}{\partial x_2} + \frac{\partial \sigma_{x_1 x_3}}{\partial x_3} &= \lambda \frac{\partial \theta}{\partial x_1} + \mu \left(\frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_2}{\partial x_2 \partial x_1} + \frac{\partial^2 u_3}{\partial x_3 \partial x_1} \right) \\ + \mu \left(\frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_1}{\partial x_2^2} + \frac{\partial^2 u_1}{\partial x_3^2} \right) &= (\lambda + \mu) \frac{\partial \theta}{\partial x_1} + \nabla^2 u_1 \end{aligned} \quad (2.7)$$

Similarly, for the left side of the second equation and third equation in (2.2), one gets,

$$\begin{aligned} \frac{\partial \sigma_{x_2 x_1}}{\partial x_1} + \frac{\partial \sigma_{x_2 x_2}}{\partial x_2} + \frac{\partial \sigma_{x_2 x_3}}{\partial x_3} &= (\lambda + \mu) \frac{\partial \theta}{\partial x_2} + \nabla^2 u_2 \\ \frac{\partial \sigma_{x_3 x_1}}{\partial x_1} + \frac{\partial \sigma_{x_3 x_2}}{\partial x_2} + \frac{\partial \sigma_{x_3 x_3}}{\partial x_3} &= (\lambda + \mu) \frac{\partial \theta}{\partial x_3} + \nabla^2 u_3 \end{aligned} \quad (2.8)$$

Finally, in terms of displacements, after some algebraic manipulations, and using the dilatation definition, the so-called field vector equation of motion is given by,

$$(\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}(x, t)) + \nabla^2 \mathbf{u}(x, t) = \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} \quad (2.9)$$

Looking ahead for finding solutions to the equation of motion (2.9), it can be written in a more convenient way using some well-known identities from vector calculus, as follows.

Recalling that

$$\varepsilon_{ijk} \varepsilon_{mnp} = \begin{vmatrix} \delta_{im} & \delta_{in} & \delta_{ip} \\ \delta_{jm} & \delta_{jn} & \delta_{jp} \\ \delta_{km} & \delta_{kn} & \delta_{kp} \end{vmatrix}, \text{ for } p = k, \quad \varepsilon_{ijk} \varepsilon_{mnk} = \begin{vmatrix} \delta_{im} & \delta_{in} & \delta_{ik} \\ \delta_{jm} & \delta_{jn} & \delta_{jk} \\ \delta_{km} & \delta_{kn} & \delta_{kk} \end{vmatrix} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}, \quad (2.10)$$

and that the curl of a vector field can be expressed in terms of Levi Civita symbol,

$$(\nabla \times \mathbf{u})_i = \varepsilon_{ijk} \partial_j u_k, \text{ then}$$

$$(\nabla \times (\nabla \times \mathbf{u}))_i = \varepsilon_{ijk} \partial_j (\nabla \times \mathbf{u})_k = \varepsilon_{ijk} \partial_j (\varepsilon_{klm} \partial_l u_m) = \varepsilon_{ijk} \varepsilon_{klm} \partial_j (\partial_l u_m), \text{ and}$$

$$(\nabla \times (\nabla \times \mathbf{u}))_i = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j \partial_l u_m = \partial_i \partial_j u_j - \partial_j \partial_j u_i = \nabla (\nabla u_j) - \nabla^2 u_i.$$

Rearranging terms, and dropping subscripts,

$$\nabla^2 \mathbf{u} = \nabla (\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u}) \quad (2.11)$$

Inserting (2.11) into (2.9), the equation of motion can be rewritten as,

$$(\lambda + 2\mu) \nabla (\nabla \cdot \mathbf{u}(x, t)) - \mu \nabla \times (\nabla \times \mathbf{u}(x, t)) = \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} \quad (2.12)$$

Now, for the solution of the equation of motion Helmholtz decomposition is applied, and the displacement field is expressed in terms of scalar and vector potentials, as given below.

$$\mathbf{u}(x, t) = \nabla \phi(x, t) + \nabla \times \Gamma(x, t) \quad (2.13)$$

Where $\nabla \times \Gamma = \nabla \times \psi + \nabla \times \nabla \times \chi$ provides for the shear wave polarization of one component in the horizontal plane, and the other component in the vertical plane. Also, recalling that the curl of a gradient and the divergence of a curl are both equal to zero, one can write,

$$\nabla \times (\nabla \phi) = 0, \quad \nabla \cdot (\nabla \times \Gamma) = 0 \quad (2.14)$$

Considering (2.14), substituting the potentials for the field displacement, and using the vector identities in (2.12), yields,

$$(\lambda + 2\mu)\nabla(\nabla^2\phi) - \mu\nabla \times \nabla \times (\nabla \times \Gamma) = \frac{\partial^2}{\partial t^2}(\nabla\phi + \nabla \times \Gamma) \quad (2.15)$$

Now, expressing $\nabla^2\mathbf{u} = \nabla(\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u})$ in terms of potentials,

$$\nabla^2(\nabla\phi + \nabla \times \Gamma) = \nabla(\nabla \cdot (\nabla\phi + \nabla \times \Gamma)) - \nabla \times (\nabla \times (\nabla\phi + \nabla \times \Gamma)).$$

And

$$\nabla^2(\nabla\phi) + \nabla^2(\nabla \times \Gamma) = \nabla^2(\nabla\phi) + \nabla(\nabla \cdot \nabla \times \Gamma) - \nabla \times (\nabla \times \nabla\phi) - \nabla \times \nabla \times (\nabla \times \Gamma).$$

But $\nabla(\nabla \cdot \nabla \times \Gamma) = 0$, $\nabla \times (\nabla \times \nabla\phi) = 0$. Then,

$$\nabla^2(\nabla \times \Gamma) = -\nabla \times \nabla \times (\nabla \times \Gamma) \quad (2.16)$$

Using Eq. (2.16), equation (2.15) simplifies to:

$$(\lambda + 2\mu)\nabla(\nabla^2\phi) + \mu\nabla^2(\nabla \times \Gamma) = \rho \frac{\partial^2}{\partial t^2}(\nabla\phi + \nabla \times \Gamma) \quad (2.17)$$

Rearranging terms by keeping all the scalar potential terms on the left side, and the vector potential on the right side, the following equality is obtained:

$$\nabla \left[(\lambda + 2\mu)\nabla^2\phi(x,t) - \rho \frac{\partial^2\phi(x,t)}{\partial t^2} \right] = -\nabla \times \left[\mu\nabla^2\Gamma(x,t) - \rho \frac{\partial^2\Gamma(x,t)}{\partial t^2} \right] \quad (2.18)$$

For the above equality (2.18) to hold, both the left side and the right side must be equal to zero.

Thus, two separate equations are obtained. The scalar equation for the compressional wave, and the vector equation for the shear wave. To wit,

$$\nabla^2\phi(x,t) = \frac{1}{c_p^2} \frac{\partial^2\phi(x,t)}{\partial t^2}, \text{ where } c_p = \sqrt{\frac{(\lambda + 2\mu)}{\rho}}, \quad (2.19)$$

where c_p is the dilatational or compressional wave velocity. And,

$$\nabla^2\Gamma(x,t) = \frac{1}{c_s^2} \frac{\partial^2\Gamma(x,t)}{\partial t^2}, \text{ where } c_s = \sqrt{\frac{\mu}{\rho}} \quad (2.20)$$

where c_s is the rotational wave velocity.

It can be verified that (2.19) and (2.20) clearly satisfy equation (2.13).

Plane Wave Solution

Rayleigh waves can be characterized as plane waves. It would be advantageous to offer a concise explanation of the key attributes related to the plane wave. This will contribute to a better understanding of the Rayleigh wave phenomenon.

Displacement fields for plane P and S waves propagating in the $x_1 - x_3$ plane containing the source and the receiver are shown in Fig. 2.1. The P-wave displacement is along the wave vector $\mathbf{k}$. The S-wave can be decomposed into two polarizations, SV and SH , perpendicular to the wave vector. The SH displacement is purely horizontal, where the SV displacement is in the $x_1 - x_3$. Wave motion is positive in the positive direction of x_1 and x_3 , as shown in Fig. 2.1.

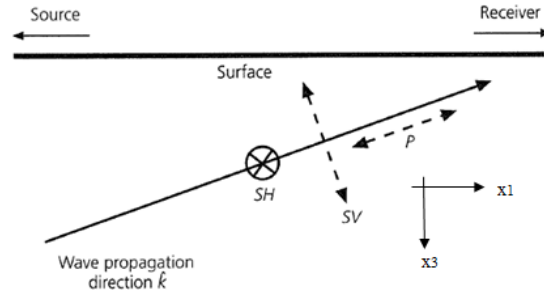


Fig. 2.1. Polarized Shear Wave

Using the scalar and vector potentials given in equation (2.13), the displacement vector can be written as follows.

$$u(x_1, x_3, t) = \nabla \phi(x_1, x_3, t) + \nabla \times \psi(x_1, x_3, t) + \nabla \times \nabla \times \chi(x_1, x_3, t) \quad (2.21)$$

where $\nabla \times \mathbf{\mathcal{G}} = \nabla \times \psi(x_1, x_3, t) + \nabla \times \nabla \times \chi(x_1, x_3, t)$, and ϕ , ψ , χ represent the scalar and vector potentials for the compressional wave P, the SV shear wave, and the SH shear wave, respectively.

It is convenient to choose the scalar potential and the vector potential such that,

$$\begin{aligned}\boldsymbol{\psi}(x_1, x_3, t) &= (0, \psi(x_1, x_3, t), 0) \\ \boldsymbol{\chi}(x_1, x_3, t) &= (0, \chi(x_1, x_3, t), 0)\end{aligned}\tag{2.22}$$

Then, each of the right-side terms in (2.21) can be expressed as follows.

$$\begin{aligned}\nabla \phi &= \left(\frac{\partial \phi}{\partial x_1}, 0, \frac{\partial \phi}{\partial x_3} \right) \\ \nabla \times \boldsymbol{\psi} &= \left(-\frac{\partial \psi}{\partial x_3}, 0, \frac{\partial \psi}{\partial x_1} \right) \\ \nabla \times \nabla \times \boldsymbol{\chi} &= \left(0, -\left[\frac{\partial^2 \chi}{\partial x_1^2} + \frac{\partial^2 \chi}{\partial x_3^2} \right], 0 \right)\end{aligned}\tag{2.23}$$

It can be observed that the P and SV waves contribute to the displacement components in the x_1 and x_3 directions, while the SH wave contribute to the x_2 displacement component only.

Grouping the displacement components yields,

$$\begin{aligned}u_1 &= \frac{\partial \phi}{\partial x_1} - \frac{\partial \psi}{\partial x_3} \\ u_3 &= \frac{\partial \phi}{\partial x_3} + \frac{\partial \psi}{\partial x_1} \\ u_2 &= -\left(\frac{\partial^2 \chi}{\partial x_1^2} + \frac{\partial^2 \chi}{\partial x_3^2} \right)\end{aligned}\tag{2.24}$$

From the above given equations, it can be noted that:

- a) P-SV waves are independent of the SH wave.
- b) For waves propagating in the x_1 - x_3 plane, the P and SV waves form a coupled system.
- c) The SH waves are decoupled from the P and SV waves.

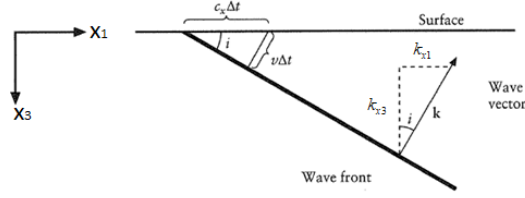


Fig. 2.2. Wave Front

It can be observed in Fig. 2.2 that the wave vector, $\mathbf{k}$, is normal to the wave front and points in the direction of propagation. For plane waves traveling in the x_1 - x_3 plane, the propagation direction is given by the wave vector with components k_{x1} and k_{x3} , or the incident angle i between the wave vector and the $-x_3$ axis. In a time-increment Δt , the wave front moves a distance $v\Delta t$, where v is the medium velocity, and sweeps out a distance along the surface $c\Delta t$, where c is the apparent velocity along the surface.

The potentials for the P and SV waves can be written,

$$\begin{aligned} (P) &\rightarrow \phi = a \exp(i(\omega t - k_{x1}x_1 - k_{x3c_p}x_3)) \\ (SV) &\rightarrow \psi = b \exp(i(\omega t - k_{x1}x_1 - k_{x3c_s}x_3)) \end{aligned} \quad (2.25)$$

Where c_p and c_s stand for the compressional and shear wave velocities of the medium, respectively.

The components of the wave vectors satisfy,

$$\begin{aligned} |k_p| &= k_{x1}^2 + k_{x3c_p}^2 = \omega^2/c_p^2 \\ |k_{sv}| &= k_{x1}^2 + k_{x3c_s}^2 = \omega^2/c_s^2 \end{aligned} \quad (2.26)$$

From Eqs. (2.26), the following expressions are obtained,

$$p = \frac{k_{x3cp}}{k_{x1}} = \left(c^2 / c_p^2 - 1 \right)^{\frac{1}{2}}$$

$$q = \frac{k_{x3cs}}{k_{x1}} = \left(c^2 / c_s^2 - 1 \right)^{\frac{1}{2}}$$
(2.27)

where $c = \frac{\omega}{k_{x1}}$.

Then the potentials for the P and SV waves can be written as follows,

$$\phi = a \exp(i(\omega t - k_{x1}x_1 - k_{x1}px_3)) = a \exp(-ik_px_3) [\exp(i(\omega t - kx_1))]$$

$$\psi = b \exp(i(\omega t - k_{x1}x_1 - k_{x1}qx_3)) = b \exp(-ik_qx_3) [\exp(i(\omega t - kx_1))]$$
(2.28)

Note that $k_{x1}p$ and $k_{x1}q$ are the wavenumbers of the pressure wave and shear wave, respectively, where $k = k_{x1}$, $k_p = kp$ and $k_q = kq$.

The equations of motion for P and SV can be written as follows,

$$\rho \frac{\partial^2 \phi}{\partial t^2} = (\lambda + 2\mu) \nabla^2 \phi, \quad \rho \frac{\partial^2 \psi}{\partial t^2} = \mu \nabla^2 \psi$$
(2.29)

where,

$$\nabla^2 = \partial^2 / \partial x_1^2 + \partial^2 / \partial x_3^2$$

Using Eqs. (2.26), and recalling that $c_p^2 = \lambda + 2\mu / \rho$ and $c_s^2 = \mu / \rho$, after some algebraic manipulations the plane wave differential equations can be written as

$$\frac{\partial^2 \phi}{\partial x_3^2} + \left(\frac{\omega^2}{c_p^2} - k^2 \right) \phi = 0$$

$$\frac{\partial^2 \psi}{\partial x_3^2} + \left(\frac{\omega^2}{c_s^2} - k^2 \right) \psi = 0$$
(2.30)

Plane wave solutions to equations (2.30) are of the following form:

$$\begin{aligned}\varphi(x, t) &= \Phi(x_3) e^{i(kx_1 - \omega t)} \\ \psi(x, t) &= \Psi(x_3) e^{i(kx_1 - \omega t)}\end{aligned}\tag{2.31}$$

Eqs. (2.31) represent a plane wave traveling in the x_1 direction. The coefficients $\Phi(x_3)$ and $\Psi(x_3)$ are time-independent functions which vary with depth.

Rayleigh Wave in Half-Space Homogeneous Elastic Medium.

Rayleigh wave propagation in an elastic homogenous medium can be formulated by considering the wave traveling in a plane, as shown in Fig.2.3.

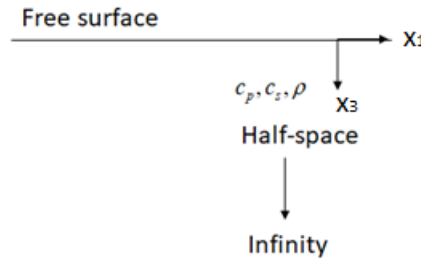


Fig. 2.3. Plane Wave Propagation in Homogeneous Medium,

For Rayleigh waves the displacement must decay as $x_3 \rightarrow \infty$. That is, for the energy to be trapped near the surface the exponentials $\exp(-ikpx_3)$, $\exp(-ikqx_3)$ in (2.28) must have real negative exponents. At the surface or free boundary, the apparent velocity c must comply with the inequality $c < c_s < c_p$. Thus, to ensure the energy to be trapped near the surface, it is required that,

$$\begin{aligned}p &= \left(\frac{c^2}{c_p^2} - 1 \right)^{1/2} = \left(-1 * \left(1 - \frac{c^2}{c_p^2} \right) \right)^{1/2} = i\bar{p} = i \left(1 - \frac{c^2}{c_p^2} \right)^{1/2} \\ q &= \left(\frac{c^2}{c_s^2} - 1 \right)^{1/2} = \left(-1 * \left(1 - \frac{c^2}{c_s^2} \right) \right)^{1/2} = i\bar{q} = i \left(1 - \frac{c^2}{c_s^2} \right)^{1/2}\end{aligned}\tag{2.32}$$

Considering wave propagation in the x_1 - x_3 plane only, Eq. (2.13) can be rewritten as given below.

$$u(x_1, x_3, t) = \nabla \phi(x_1, x_3, t) + \nabla \times \psi(x_1, x_3, t) \quad (2.33)$$

Now, inserting Eqs. (2.28) in Eqs. (2.33), the scalar and vector potentials for plane wave harmonic solutions can be obtained as follows.

$$\phi = [a_1 \exp(-k\bar{p}x_3)] e^{i(\omega t - kx_1)}, \quad \psi = [b_1 \exp(-k\bar{q}x_3)] e^{i(\omega t - kx_1)} \quad (2.34)$$

and the displacements are:

$$\begin{aligned} u_1 &= \frac{\partial \phi}{\partial x_1} - \frac{\partial \psi}{\partial x_3} = (-a_1 i k \exp(-k\bar{p}x_3) + b_1 k \bar{q} \exp(-k\bar{q}x_3)) e^{i(\omega t - kx_1)} \\ u_3 &= \frac{\partial \phi}{\partial x_3} + \frac{\partial \psi}{\partial x_1} = (-a_1 k \bar{p} \exp(-k\bar{p}x_3) - b_1 i k \exp(-k\bar{q}x_3)) e^{i(\omega t - kx_1)} \end{aligned} \quad (2.35)$$

Similarly, the components of the symmetric stress tensor can be expressed in terms of the potentials and the harmonic functions, as shown below.

$$\begin{aligned} \tau_{31} &= \mu \left(2 \frac{\partial^2 \phi}{\partial x_1 \partial x_3} + \frac{\partial^2 \psi}{\partial x_1^2} - \frac{\partial^2 \psi}{\partial x_3^2} \right) = \\ &\mu [a_1 2k^2 \bar{p} \exp(-k\bar{p}x_3) + b_1 k^2 \exp(-k\bar{q}x_3)(1 - \bar{q}^2)] e^{i(\omega t - kx_1)} \\ \tau_{33} &= \lambda \left(\frac{\partial^2 \phi}{\partial x_1^2} + \frac{\partial^2 \phi}{\partial x_3^2} \right) + 2\mu \left(\frac{\partial^2 \phi}{\partial x_3^2} + \frac{\partial^2 \psi}{\partial x_1 \partial x_3} \right) = \\ &\left[(a_1 k^2 \exp(-k\bar{p}x_3)(\lambda + \lambda \bar{p}^2 + 2\mu \bar{p}^2) + b_1 2\mu k^2 \bar{q} \exp(-k\bar{q}x_3)) \right] e^{i(\omega t - kx_1)} \end{aligned} \quad (2.36)$$

Boundary Conditions.

At the surface or free boundary, where $x_3 = 0$, the stresses vanish. Hence,

$$\begin{aligned} \tau_{31} &= 2\bar{p}a_1 + (1 - \bar{q}^2)b_1 = 0 \\ \tau_{33} &= [\lambda(1 + \bar{p}^2) + 2\mu\bar{p}^2]a_1 + 2\mu\bar{q}b_1 = 0 \end{aligned} \quad (2.37)$$

or

Using Eq. (2.32), replacing Lamé's parameters,

$$c_p^2 = \frac{\lambda + 2\mu}{\rho}, \quad c_s^2 = \frac{\mu}{\rho}, \quad \mu = \rho c_s^2, \quad \lambda = \rho c_p^2 - 2\mu = \rho c_p^2 - 2\rho c_s^2,$$

and rearranging terms,

$$\begin{aligned} \tau_{31} &= 2\bar{p}a_1 + (1 - \bar{q}^2)b_1 = 2i\left(1 - \frac{c^2}{c_p^2}\right)^{1/2} a_1 + \left(2 - \frac{c^2}{c_s^2}\right)b_1 = 0 \\ \tau_{33} &= \left[\lambda(1 + \bar{p}^2) + 2\mu\bar{p}^2\right]a_1 + 2\mu\bar{q}b_1 = \left[\lambda\left(\frac{c^2}{c_p^2}\right) + 2\mu\left(\frac{c^2}{c_p^2}\right)\right]a_1 + 2\mu i\left(1 - \frac{c^2}{c_s^2}\right)^{1/2} b_1 = 0 \\ \left[c_p^2\left(\frac{c^2}{c_p^2}\right) - 2c_s^2\left(\frac{c^2}{c_p^2}\right) + 2c_s^2\left(\frac{c^2}{c_p^2} - 1\right)\right]a_1 + 2c_s^2 i\left(1 - \frac{c^2}{c_s^2}\right)^{1/2} b_1 &= 0 \\ -\left(2 - \frac{c^2}{c_s^2}\right)a_1 + 2i\left(1 - \frac{c^2}{c_s^2}\right)^{1/2} b_1 &= 0 \end{aligned} \tag{2.38}$$

The equation for τ_{31} in (2.37) can be written as,

$$2i\left(1 - \frac{c^2}{c_p^2}\right)^{1/2} a_1 + \left(2 - \frac{c^2}{c_s^2}\right)b_1 = 0$$

Similarly, dropping the density ρ , the equation for τ_{33} can be written as

$$-\left(2 - \frac{c^2}{c_s^2}\right)a_1 + 2i\left(1 - \frac{c^2}{c_s^2}\right)^{1/2} b_1 = 0$$

Finally, the following system of equations is obtained:

$$\begin{aligned} 2i\left(1 - \frac{c^2}{c_p^2}\right)^{1/2} a_1 + \left(2 - \frac{c^2}{c_s^2}\right)b_1 &= 0 \\ -\left(2 - \frac{c^2}{c_s^2}\right)a_1 + 2i\left(1 - \frac{c^2}{c_s^2}\right)^{1/2} b_1 &= 0 \end{aligned} \tag{2.39}$$

$$\begin{vmatrix} 2\gamma_\alpha & i(2 - c^2/\beta^2) \\ i(2 - c^2/\beta^2) & -2\gamma_\beta \end{vmatrix} = 0,$$

$$\left(2 - \frac{c^2}{\beta^2}\right)^2 - 4\gamma_\alpha\gamma_\beta = 0.$$

For the solution of the above system of equations, it is required to solve the eigenvalue problem given by,

$$\begin{vmatrix} 2i(c^2/c_p^2 - 1)^{1/2} & (2 - c^2/c_s^2) \\ -(2 - c^2/c_s^2) & 2i(c^2/c_s^2 - 1)^{1/2} \end{vmatrix} = 0 \quad (2.40)$$

The corresponding frequency or period equation can be derived as follows.

$$\left(2 - c^2/c_s^2\right)^2 - 4\left(c^2/c_p^2 - 1\right)^{1/2}\left(c^2/c_s^2 - 1\right)^{1/2} = 0 \quad (2.41)$$

Eq. (2.41) is known as Rayleigh Period Equation. This equation can be expressed in a more rational way by performing the following reformulation:

Write the displacement vector field,

$$\mathbf{u} = \left[A(\mathbf{e}_1 - i\gamma_p\mathbf{e}_3)e^{-\gamma_p k x_3} + B(i\gamma_s\mathbf{e}_1 + \mathbf{e}_3)e^{-\gamma_s k x_3} \right] e^{i(\omega t - k x_1)} \quad (2.42)$$

where,

$k = \omega/c$, $\gamma_p = \left(1 - c^2/c_p^2\right)^{1/2}$, $\gamma_s = \left(1 - c^2/c_s^2\right)^{1/2}$, and c_p , c_s are the compressional and shear wave phase velocities, respectively.

Imposing boundary conditions at $x_3 = 0$, where the stresses vanish, recalling that,

$$\tau_{31} = \mu \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right), \text{ and } \tau_{33} = \lambda \left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_3}{\partial x_3} \right) + 2\mu \frac{\partial u_3}{\partial x_3},$$

and using the well-known relationships between the Lamé coefficients and the body wave phase velocities, yields,

$$\begin{aligned} 2\gamma_p A + i \left(2 - \frac{c^2}{c_s^2} \right) B &= 0 \\ -i \left(2 - \frac{c^2}{c_s^2} \right) A + 2\gamma_s B &= 0 \end{aligned} \quad (2.43)$$

From Eq. (1.2) the Rayleigh Period Equation can be derived as follows.

$$\frac{A}{B} = - \frac{i \left(2 - \frac{c^2}{c_s^2} \right)}{2\gamma_p} = \frac{2\gamma_s}{i \left(2 - \frac{c^2}{c_s^2} \right)}, \text{ and finally, one writes,}$$

$$\left(2 - c^2/c_s^2 \right)^2 - 4\gamma_p\gamma_s = 0 \quad (2.44)$$

This is the well-known “Raleigh Period Equation”.

Now, expanding Eq. (2.44) and substituting for γ_p and γ_s , Eq. (1.3) can be rewritten as,

$$\left(2 - c^2/c_s^2 \right)^2 = 4 \left(1 - \frac{c^2}{c_p^2} \right)^{\frac{1}{2}} \left(1 - \frac{c^2}{c_s^2} \right)^{\frac{1}{2}}$$

Defining $c_p = \kappa c_s$, where $\kappa = \sqrt{\left(\frac{2(-1+\nu)}{2\nu-1} \right)}$, ν is the Poisson ratio, and $\xi = c^2/c_s^2$, one writes,

$$4 + \xi^2 - 4\xi = 4 \sqrt{1 - \frac{\xi}{\kappa^2}} \sqrt{1 - \xi} \quad (2.45)$$

The Rayleigh Period Equation can be written as given in Eq. (2.45), but with the right-side term moved to the left side as follows.

$$4 + \xi^2 - 4\xi - 4 \sqrt{1 - \frac{\xi}{\kappa^2}} \sqrt{1 - \xi} = 0 \quad (2.46)$$

To derive the Rayleigh equation, as is currently known, Eq. (2.46) is rationalized, rearranging terms, and defining the variable $\xi = \frac{c^2}{c_s^2}$, yields,

$$\xi^3 - 8\xi^2 + 8\xi \left(3 - 2\frac{c_s^2}{c_p^2} \right) - 16 \left(1 - \frac{c_s^2}{c_p^2} \right) = 0 \quad (2.47)$$

Equation (2.47) is the equation as originally derived by Lord Rayleigh (1845), which has been reproduced in many books and publications, (e.g., Ewing, W. M., et al., 1957; Achenbach, J.D., 1973; Stein, S., and Wysession, M., 1979; Ben-Menahem, A., and Singh, S. J., 1980; Pujol, J., 2003).

It is known that Eq. (2.47) has a solution $0 < c_R < 1$, subject to the condition:

$$c_R < c_s < c_p \quad (2.48)$$

For a Poisson solid, $\nu = 0.25$ and $\kappa = 1.73$. Then, in addition to $\xi = 0$, the roots of Eq. (2.47) are:

$$\xi = 0.845, \xi = 3.141, \xi = 4.014.$$

Among these solutions, only one complies with the condition $c < c_s < c_p$. That is,

$$\xi = 0.845, c = c_R = 0.919c_s$$

The Rayleigh Period Equation (2.46) for convenience is repeated below.

$$4 + \xi^2 - 4\xi - 4\sqrt{1 - \frac{\xi}{\kappa^2}}\sqrt{1 - \xi} = 0$$

Plugging in Eq. (2.46) the solution of the Rayleigh equation (2.47) with $\kappa = 1.73$ and $\xi = 0.845$, yields,

$$\xi = 0$$

That is, for $c_R < c_s$ the roots of the rationalized Rayleigh equation also satisfy the Rayleigh Period Equation. However, this is not true either for real roots $c_R > c_s$ or complex roots.

Table 2.1 given below, shows the roots for several Poisson ratios of both the Rayleigh Period Equation and the rationalized Rayleigh equation. It can be observed that the roots of the Rayleigh

Period Equation exactly match the roots of the traditional Rayleigh equation for $c < c_s < c_p$. The other roots of the traditional Rayleigh equation are results of algebraic manipulations. These roots are not intrinsic to the Rayleigh Period Equation.

Poisson Ratio	Rayleigh Period Equation		Rayleigh Rationalized Equation		
ν	Root 1	Root 2	Root 1	Root 2	Root 3
0.20	0.0000	0.8299	0.8299	2.6889	4.4812
0.25	0.0000	0.8453	0.8453	3.1547	4.0000
0.30	0.0000	0.8601	0.8601	$3.5700 - 0.7369 i$	$3.5700 + 0.7369 i$
0.40	0.0000	0.8877	0.8877	$3.5561 - 0.5406 i$	$3.5561 + 0.5406 i$
0.49	0.0000	0.9103	0.9103	$3.5700 - 0.7369 i$	$3.5700 + 0.7369 i$

Table 2.1. Roots of Rayleigh Period Equation and Traditional Rayleigh Equation

Approximate Solutions of Rayleigh Equation

Analytical formulas exist to approximate the Rayleigh wave velocity as a function of Poisson ratios, other than the one for a Poisson solid, $\nu = 0.25$. Eq. (2.41) and Eq. (2.42) show Achenbach and Bergman formulas, respectively.

$$\frac{c_R}{c_s} = \frac{0.862 + 1.14\nu}{1 + \nu} \quad (2.49)$$

$$\frac{c_R}{c_s} = \frac{0.87 + 1.12\nu}{1 + \nu} \quad (2.50)$$

Particle Motion Kinematic at Surface

For the sake of simplicity, a Poisson solid is considered. For a Poisson solid the displacements at the surface are obtained as follows.

$$\begin{aligned}
2\left(c^2/c_p^2 - 1\right)^{1/2} a_1 + (2 - c^2/c_s^2) b_1 &= 0 \\
\left(c^2/c_s^2 - 2\right) a_1 + 2\left(c^2/c_s^2 - 1\right)^{1/2} b_1 &= 0 \\
b_1 = -\frac{\left(c^2/c_s^2 - 2\right)}{2\left(c^2/c_s^2 - 1\right)^{1/2}} a_1 &= \frac{\left(2 - c^2/c_s^2\right)}{-2i\left(1 - c^2/c_s^2\right)^{1/2}} a_1 = \frac{i\left(2 - c^2/c_s^2\right)}{2\left(1 - c^2/c_s^2\right)^{1/2}} a_1
\end{aligned} \quad (2.51)$$

Using Eqs (2.35), expanding and discarding the imaginary part, the particle displacement equations for a Poisson solid are:

$$\begin{aligned} i\bar{p} &= i \left(1 - \frac{c^2}{c_p^2} \right)^{1/2} = i \left(1 - \frac{0.845c_s^2}{3c_s^2} \right)^{1/2} = 0.85i \\ i\bar{q} &= i \left(1 - \frac{c^2}{c_s^2} \right)^{1/2} = i \left(1 - \frac{0.845c_s^2}{c_s^2} \right)^{1/2} = 0.39i \end{aligned}$$

$$e^{i(\omega t - kx_1)} = [\cos(\omega t - kx_1) + i \sin(\omega t - kx_1)]$$

$$\phi = [a_1 \exp(-k\bar{p}x_3)] e^{i(kx_1 - \omega t)} = a_1 \exp(-0.85kx_3) [\cos(\omega t - kx_1) + i \sin(\omega t - kx_1)]$$

$$\psi = \left[\frac{i(2 - c^2/c_s^2)}{2(1 - c^2/c_s^2)^{1/2}} a_1 \exp(-k\bar{q}x_3) \right] e^{i(kx_1 - \omega t)} = i1.48a_1 \exp(-0.39kx_3) [\cos(\omega t - kx_1) + i \sin(\omega t - kx_1)]$$

$$u_1 = \frac{\partial \phi}{\partial x_1} + \frac{\partial \psi}{\partial x_3} = k [a_1 \exp(-0.85kx_3) - 0.58a_1 \exp(-0.39kx_3)] \sin(\omega t - kx_1) \quad (2.52)$$

$$u_3 = \frac{\partial \phi}{\partial x_3} - \frac{\partial \psi}{\partial x_1} = k [-0.85a_1 \exp(-0.85kx_3) + 1.48a_1 \exp(-0.39kx_3)] \cos(\omega t - kx_1)$$

At surface, $x_3 = 0$. Then, noting that the vertical and horizontal displacements are 90° out of phase, discarding the imaginary part the equations for the displacement components can be written as

$$\begin{aligned} u_1 &= ka_1 [1 - 0.58] \sin(\omega t - kx_1) = 0.42ka_1 \sin(\omega t - kx_1) \\ u_3 &= ka_1 [-0.85 + 1.48] \cos(\omega t - kx_1) = 0.63ka_1 \cos(\omega t - kx_1) \\ \frac{u_1}{(0.42)} &= ka_1 \sin(kx_1 - \omega t) \quad (2.53) \\ \frac{u_3}{(0.63)} &= ka_1 \cos(kx_1 - \omega t) \end{aligned}$$

Finally, the geometry of the kinematic of the particle wave motion is given by the following equation (2.54), which shows that the Rayleigh wave particle motion is elliptical.)

$$\frac{u_1^2}{[ka_1(0.42)]^2} + \frac{u_3^2}{[ka_1(0.63)]^2} = 1 \quad (2.54)$$

Direction of Wave Particle Motion.

Consider

$$\begin{aligned} u_1 &= 0.42ka_1 \sin(\omega t) \\ u_3 &= 0.63ka_1 \cos(\omega t) \end{aligned} \quad (2.55)$$

From Eq. (2.28) and Fig.2.4, it can be observed that for $t = 0$,

$$\begin{aligned} u_1 &= 0.42ka_1 \sin(\omega t) = 0 \\ u_3 &= 0.63ka_1 \cos(\omega t) = 0.63ka_1 \end{aligned}$$

As t increases the particle moves to the right and up, that is, that particle moves describing a retrograde elliptical trajectory. At a given time the maximum ratio $u_3/u_1 = 1.5$.

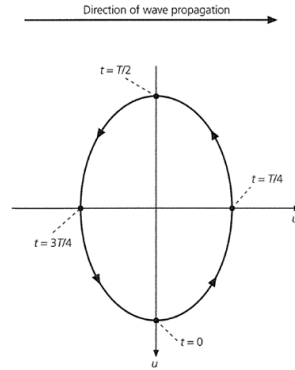


Fig. 2.4. Retrograde Elliptical Particle Motion

Prograde Particle Motion

It is broadly known that Rayleigh wave particle motion is elliptical retrograde at the surface or free boundary for real roots of the equation of motion. Also, the particle motion becomes prograde at a depth of about one-fifth of the wavelength. In this section it is mathematically shown the depth at which the displacement component, u_1 , becomes equation zero, and also when it becomes negative, which indicates the change from retrograde to prograde particle motion.

As given before, the vector equation of motion can be written as follows.

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla \nabla \cdot \mathbf{u} = \rho \ddot{\mathbf{u}} \quad (2.56)$$

A time-harmonic plane wave solution for the u_1 displacement component is given by (Pujols, 2003):

$$u_1 = \frac{i\gamma_{cs}}{(1 - c^2/2c_s^2)} b_1 \left[-e^{-\gamma_{cp} k x_3} + \left(1 - \frac{c^2}{2c_s^2} \right) e^{-\gamma_{cs} k x_3} \right] e^{i(\omega t - k x_1)} \quad (2.57)$$

And

$$u_3 = \frac{\gamma_{cp} \gamma_{cs}}{(1 - c^2/2c_s^2)} b_1 \left[-e^{-\gamma_{cp} k x_3} + \left(1 - \frac{c^2}{2c_s^2} \right) e^{-\gamma_{cs} k x_3} \right] e^{i(\omega t - k x_1)} \quad (2.58)$$

where,

$$\gamma_{cp} = \left(1 - \frac{c^2}{c_p^2} \right)^{\frac{1}{2}}, \quad \gamma_{cs} = \left(1 - \frac{c^2}{c_s^2} \right)^{\frac{1}{2}}$$

It is of interest to examine for which value of x_3 the displacement component $u_1 = 0$. That is,

$$-e^{-\gamma_{cp} k x_3} + \left(1 - \frac{c^2}{2c_s^2} \right) e^{-\gamma_{cs} k x_3} = 0, \text{ or in dimensionless form:}$$

$$\begin{aligned} -e^{-\left(1 - \frac{\xi^2}{\kappa^2}\right)^{\frac{1}{2}} \zeta} + \left(1 - \frac{\xi^2}{2} \right) e^{-\left(1 - \xi^2\right)^{\frac{1}{2}} \zeta} &= 0 \\ e^{\left[\left(1 - \xi^2\right)^{\frac{1}{2}} - \left(1 - \frac{\xi^2}{\kappa^2}\right)^{\frac{1}{2}} \right] \zeta} &= \left(1 - \frac{\xi^2}{2} \right) \end{aligned} \quad (2.59)$$

$$\kappa^2 = c_p^2 / c_s^2, \quad \xi^2 = c^2 / c_s^2, \quad \zeta = k x_3 = \frac{2\pi}{\lambda} x_3,$$

λ = wavelength.

For a Poisson solid:

$$\kappa^2 = 3, \xi^2 = 0.85$$

Solving Eq. (2.59) yields,

$$\zeta = \frac{2\pi}{\lambda} x_3 = 1.21$$

$$x_3 = 0.193\lambda$$

Thus, $u_1 = 0$ at depth approximately equal to one-fifth of the wavelength, and the motion is purely vertical.

Now, it is of interest to examine the behavior of the u_1 displacement component beyond the depth where it becomes equal to zero.

Since at given x_1 and t , u_1 depends only on x_3 , and dropping the exponential term $e^{i(\omega t - kx_1)}$, Eq. (2.57) can be written as,

$$u_1 = \frac{i\gamma_{cs}}{(1 - c^2/2c_s^2)} b_1 \left[-e^{-\gamma_{cp} k x_3} + \left(1 - \frac{c^2}{2c_s^2} \right) e^{-\gamma_{cs} k x_3} \right]$$

$$u_1 = b_1 \left[-e^{-\left(1 - \frac{\xi^2}{\kappa^2} \right) \zeta} + \left(1 - \frac{\xi^2}{2} \right) e^{-(1 - \xi^2) \zeta} \right] \quad (2.60)$$

$$b_1 = 1$$

$$u_1 = \left[-e^{-\left(1 - \frac{\xi^2}{\kappa^2} \right) \zeta} + \left(1 - \frac{\xi^2}{2} \right) e^{-(1 - \xi^2) \zeta} \right]$$

The graphs below shows that u_1 becomes negative beyond certain depth for the Poisson ratios given in Table 2.1. Thus, the particle motion changes from retrograde to prograde (Achenbach, 1973).

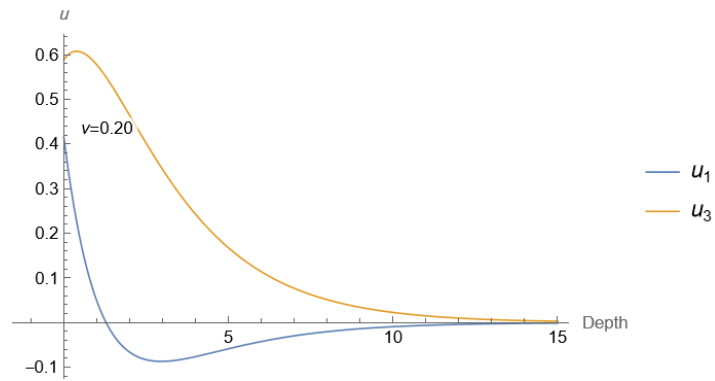


Fig. 2.5. Prograde Particle Motion at Depth $> 0.20 \lambda$ for $v=0.20$

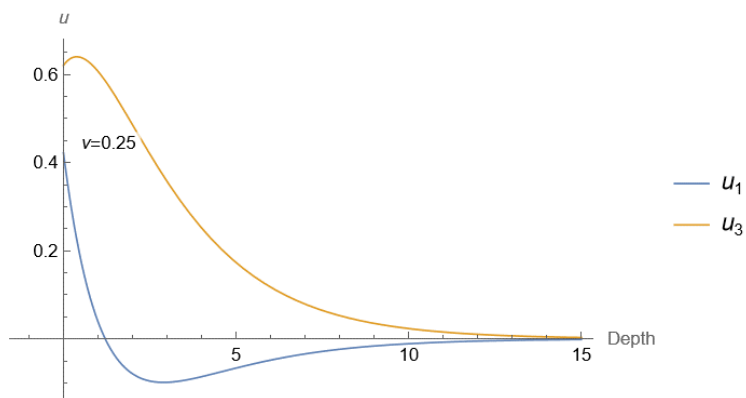


Fig. 2.6. Prograde Particle Motion at Depth $> 0.20 \lambda$ for $v=0.25$

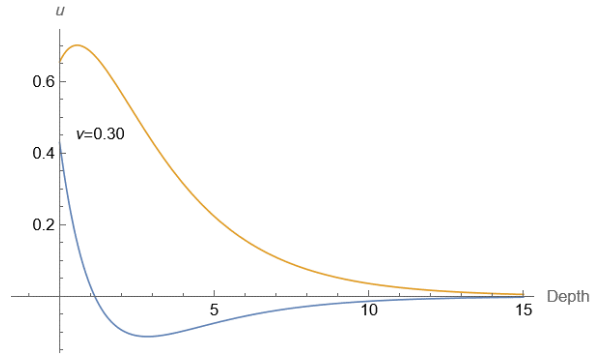


Fig. 2.7. Prograde Particle Motion at Depth $> 0.20 \lambda$ for $v=0.30$

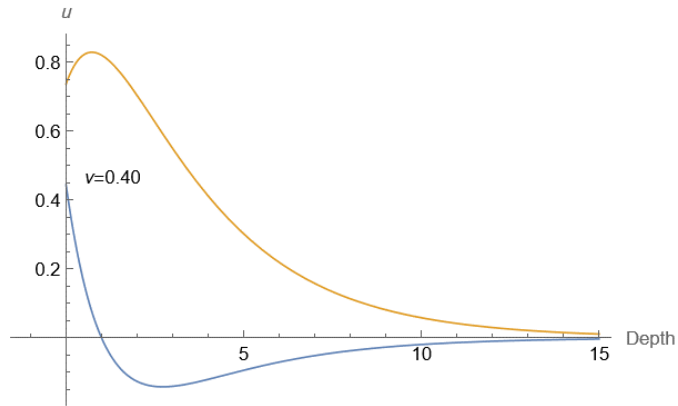


Fig. 2.8. Prograde Particle Motion at Depth $> 0.20 \lambda$ for $v=0.40$

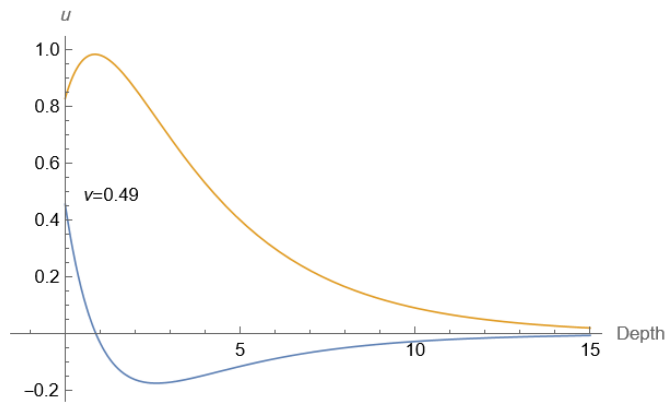


Fig. 2.9. Prograde Particle Motion at Depth $> 0.20 \lambda$ for $v=0.49$

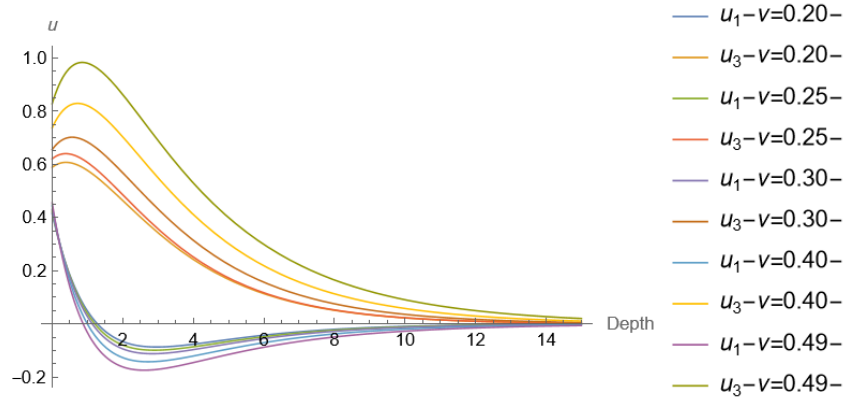


Fig. 2.10. Displacement and Direction of Motion Trends

These results show a change in sign of the horizontal displacement component, implying a change in the particle motion from retrograde to prograde at depths beyond 0.20λ . That is, defining ellipticity as the ratio of $H/V = u_1/u_3$, it can be said that a change of sign of the ellipticity means a change of the motion direction either from retrograde to prograde, or conversely from prograde to retrograde. In Fig. 2.10, it can be observed a certain trend of the horizontal displacement component changing sign at depths closer to the surface with increasing Poisson ratio values, while the magnitude of the vertical component increases as the Poisson ratio value increases. Interestingly, the displacement vertical component reaches a maximum at the depth where the particle motion changes from retrograde to prograde. Also, it can be noted that the ellipticity, H/V , is always positive at ground surface.

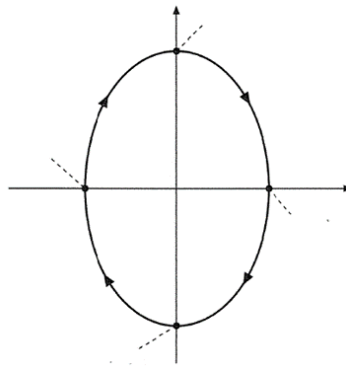


Fig. 2.11. Prograde Elliptical Particle Motion

Rayleigh Wave Ellipticity at Free Boundary.

At the free boundary the ellipticity is formulated as follows.

$$\frac{H}{V} = \frac{u_1}{u_3} = \frac{(b_1 \bar{q} - a_1 i)}{(-a_1 \bar{p} - b_1 i)} \quad (2.61)$$

From (2.37),

$$b_1 = -2 \frac{\bar{p} a_1}{(1 - \bar{q}^2)} \quad (2.62)$$

Inserting (2.47) into (2.46) yields,

$$\frac{H}{V} = \frac{u_1}{u_3} = \frac{\left(2 \frac{\left(1 - \frac{c^2}{c_p^2} \right)^{1/2}}{\left(-\frac{c^2}{c_s^2} \right)} \left(1 - \frac{c^2}{c_s^2} \right)^{1/2} - i \right)}{\left(i \left(1 - \frac{c^2}{c_p^2} \right)^{1/2} + 2 \frac{\left(1 - \frac{c^2}{c_p^2} \right)^{1/2}}{\left(-\frac{c^2}{c_s^2} \right)} \right)} \quad (2.63)$$

After some manipulation, and applying perturbation methods, Eq. (2.48) can be reduced to [1]:

$$\frac{H}{V} = \sqrt{p/q} = 2 \frac{\sqrt{1 - c_R^2/c_s^2}}{2 - c_R^2/c_s^2} \quad (2.64)$$

Now, plugging Eq. (2.42) into Eq. (2.49), yields,

$$\frac{H}{V} = 2 \frac{\sqrt{1 - \left(\frac{0.87 + 1.12\nu}{1 + \nu} \right)^2}}{2 - \left(\frac{0.87 + 1.12\nu}{1 + \nu} \right)^2} \quad (2.65)$$

The Poisson ratio, ν , can have values of $-1 \leq \nu \leq 0.50$. Inputting values of ν from 0 to 0.50 in Eq. (2.64) shows that the ellipticity is a positive function of the Poisson ratio, as shown in the figure below for $0 \leq \nu \leq 0.50$.

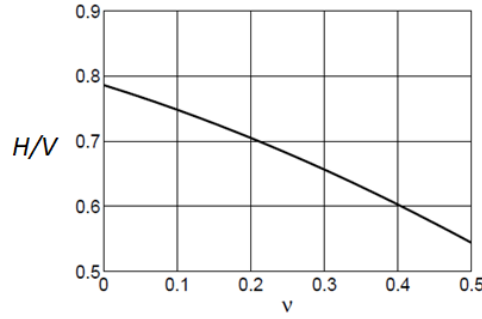


Fig. 2.12. Ellipticity vs ν

It can be concluded that at the free boundary, in homogeneous medium, the ellipticity is always positive.

Conclusions

In this chapter of the thesis, a brief introduction to the field equation of motion in an elastic, isotropic, and homogeneous medium is presented, followed by the derivation of the Rayleigh Period equation. The roots of the Rayleigh Period Equation are then compared with the roots of the traditional Rationalized Rayleigh Equation. It is demonstrated that for a range of Poisson ratios that includes most available materials, there is no need for the Rationalized Rayleigh Equation to find the real roots of the Rayleigh Period Equation. Furthermore, it is shown that the real roots of the Rationalized Rayleigh equation, which represent phase velocities larger than the shear wave velocity of the medium, are not inherent to the Rayleigh Period Equation. The same can be said for the complex roots, as both types of roots are merely a byproduct of algebraic manipulations of the Rayleigh Period Equation.

This chapter also sheds light on the kinematics of wave particle motion at the free boundary, a topic that is seldom addressed in most publications related to Rayleigh waves. The equation for the elliptical orbit of particle motion at the medium surface is derived. It is also demonstrated that the ellipticity, H/V , defined as the ratio of the horizontal component of particle displacement at the surface to the vertical component of particle displacement, regardless the Poisson ratio, is always positive at the free boundary, representing retrograde particle motion. However, it is shown that the particle motion changes from retrograde to prograde at depths equal to or beyond one-fifth of the wavelength. The point at which the particle motion changes from retrograde to prograde gradually shifts towards the free boundary as the Poisson ratio of medium increases.